

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 2 Mathematics

Assessment 2

June 2014

TIME ALLOWED: 75 minutes

Instructions:

- ***Start each question on a new page.***
- Write your name and class at the top of this page, and on your answer booklet.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- Write in blue or black pen only.
- It is suggested that you spend no more than 7 minutes on Part A.
- Approved calculators may be used.
- Standard Integrals are supplied at the rear of this paper. This is the only sheet which may be detached from any booklet.

PART A: (5 Marks)

Answers to these multiple choice should be completed on the multiple choice answer sheet supplied with your answer booklet.

All questions are worth 1 mark

1	A Primitive of $\frac{1}{\sqrt{x^2+16}}$ is A. $\frac{1}{4}\ln(x^2 + 16)$ B. $\frac{1}{2}\sqrt{x^2 + 16}$ C. $\ln(x + \sqrt{x^2 + 16})$ D. $\frac{1}{4}\tan^{-1}(x^2 + 16)$
2	$P(x)$ is a monic polynomial with real coefficients and has zeros of $2 + i$, 2 and -2. $P(x) =$ A. $x^4 - 4x^3 + x^2 + 16x - 20$ B. $x^4 + 4x^3 + x^2 - 16x - 20$ C. $x^4 + 2ix^3 + 9x^2 - 8ix + 20$ D. $x^3 - (2 + i)x^2 - 4x + (8 + 4i)$
3	$\int \sec^2 x \tan x dx =$ A. $\tan x + k$ B. $\tan^2 x + k$ C. $\frac{1}{2}\tan^2 x + k$ D. $\frac{1}{3}\sec^3 x + k$

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If two of its roots are equal, the roots of $2x^3 + 3x^2 - 12x + 7 = 0$ are:

A. $x = -1$ and $x = \frac{7}{2}$

B. $x = 1$ and $x = -\frac{7}{2}$

C. $x = -2$ and $x = -\frac{7}{8}$

D. $x = 2$ and $x = \frac{7}{8}$

5

$$\int \sin^3 x \, dx =$$

A. $\frac{1}{4}\sin^4 x + k$

B. $\frac{1}{4}\cos^4 x + k$

C. $\frac{1}{3}\cos^3 x - \cos x + k$

D. $-\cos x - \frac{1}{3}\cos^3 x + k$

PART B

(START EACH QUESTION ON A NEW PAGE)

QUESTION 6: (10 Marks)

Marks

1 (a) Find $\int \frac{dx}{1 - \sin^2 x}$

2 (b) Evaluate $\int_2^7 \frac{x-2}{\sqrt{x+2}} dx$

2 (c) (i) Find values of a, b and c, so that

$$\frac{5x^2 - x - 2}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

2 (ii) Hence find $\int \frac{5x^2 - x - 2}{(x+1)(x^2+1)} dx$

3 (d) Sketch the curve, $y = \ln|\sin x|$ for $-2\pi \leq x \leq 2\pi$, showing all keypoints

QUESTION 7: (10 Marks) *(Start on a new page)*

Marks

- 3 (a) Using t-results, or otherwise, evaluate:

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$$

- 3 (b) The equation $ax^4 + bx^3 + cx + d = 0$ has a triple root.

Show that $4a^2c + b^3 = 0$

- 1 (c) (i) Solve the equation $\cos 5\theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$

- 3 (ii) Using the substitution $x = \cos \theta$,

solve $16x^5 - 20x^3 + 5x + 1 = 0$ and hence show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$

(You may assume the result: $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$)

QUESTION 8: (10 Marks) *(Start on a new page)*

Marks

- 5 (a) By using the substitution $x = 3\sin \theta$, or otherwise, show that

$$\int \frac{x^2}{\sqrt{9-x^2}} dx = \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9-x^2} + k$$

- (b) The real roots of $x^3 + 4x - m = 0$ are α , β and γ .

1 (i) Find the value of $\frac{1}{\alpha^2\beta\gamma} + \frac{1}{\alpha\beta^2\gamma} + \frac{1}{\alpha\beta\gamma^2}$

1 (ii) Explain why $\frac{1}{\alpha^2\beta\gamma} = \frac{1}{m\alpha}$

- 3 (iii) Hence, or otherwise, find the cubic polynomial whose roots are

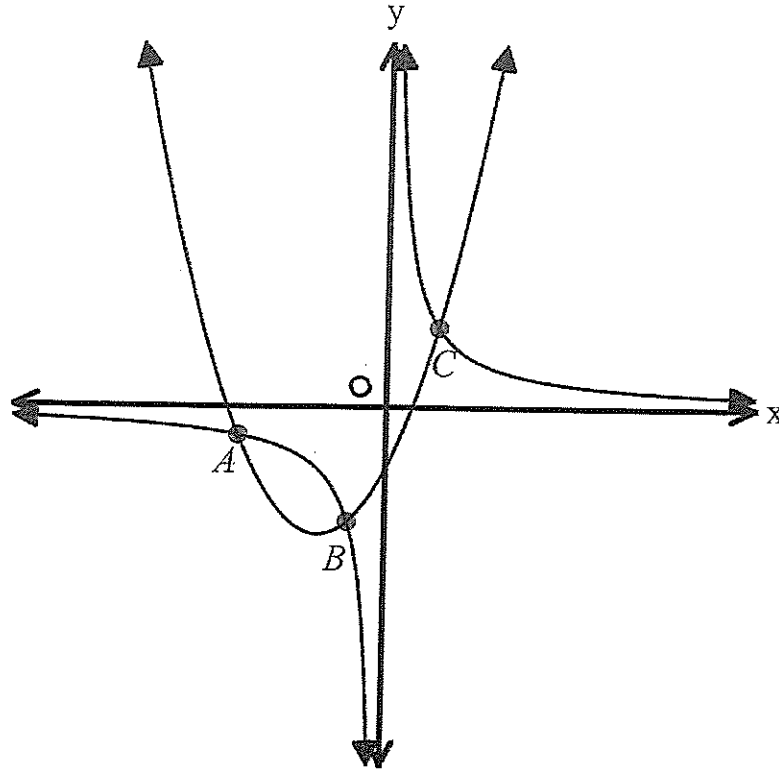
$$\frac{1}{\alpha^2\beta\gamma}, \frac{1}{\alpha\beta^2\gamma}, \text{ and } \frac{1}{\alpha\beta\gamma^2}$$

QUESTION 9: (10 Marks) *(Start on a new page)*

Marks

2 (a) Find $\int \frac{dx}{\sqrt{3+2x-x^2}}$

(b)



In the diagram above, the points A, B and C represent the points of intersection of the curve $y = x^2 + 2x - 1$ and the curve $y = \frac{1}{x}$. O is the origin

The x-values of A, B and C are α , β , and γ .

- 1 (i) Show that α , β , and γ satisfy $x^3 + 2x^2 - x - 1 = 0$
- 2 (ii) Find a polynomial with roots α^2 , β^2 and γ^2
- 2 (iii) Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$
- 3 (iv) Prove that $OA^2 + OB^2 + OC^2 = 11$

SOLUTIONS - YEAR 12 TERM 2

EXTENSION 2 ASSESSMENT

Multiple Choice

1/ C 2/ A 3/ C 4/ B 5/ C.

QUESTION 6:

$$(a) \int \frac{dx}{1 - \sin^2 x} = \int \frac{dx}{\cos^2 x} = \tan x + k$$

$$(b) \int_2^7 \frac{x+2}{\sqrt{x+2}} dx = \int_2^7 \frac{4}{\sqrt{x+2}} dx$$

$$= \left[\frac{2}{3} (x+2)^{3/2} \right]_2^7 - \left[8 (x+2)^{1/2} \right]_2^7$$

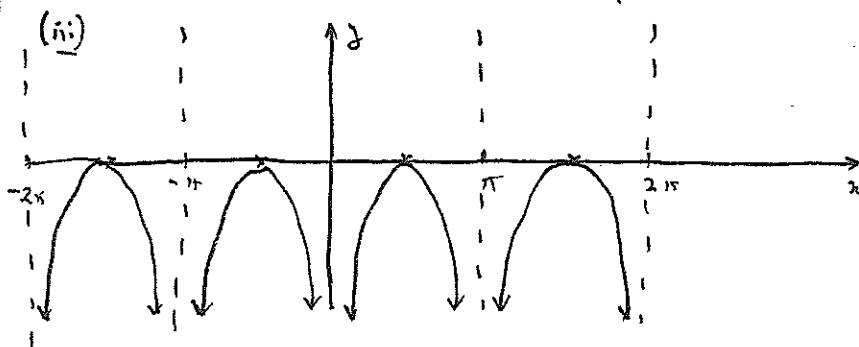
$$= 18 - \frac{16}{3} - 8(3-2)$$

$$= \frac{14}{3}$$

(c) (i) $a = 2, b = 3, c = -4$

$$(ii) \int \frac{5x^2 - x - 2}{(x+1)(x^2+1)} dx = \int \frac{2}{x+1} dx + \int \frac{3x-4}{(x^2+1)} dx$$

$$= 2 \ln(x+1) + \frac{3}{2} \ln(x^2+1) - 4 \tan^{-1} x + k$$



QUESTION 7:

$$t = \tan^2 \frac{x}{2} \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$x = 0, t = 0$$

$$x = \frac{\pi}{2}, t = 1$$

$$\therefore dx = \frac{2}{\sec^2 \frac{x}{2}} dt$$

$$= \frac{2}{1+t^2} dt$$

$$\therefore \int_0^1 \frac{2}{1+t^2} \frac{dt}{2 + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2 dt}{2 + 2t^2 + 1 - t^2}$$

$$= 2 \int_0^1 \frac{dt}{t^2 + 3}$$

$$= 2 \cdot \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$(b) \quad P'(x) = 4ax^3 + 3bx^2 + c$$

$$P''(x) = 12ax^2 + 6bx$$

For triple roots, $P'(x) = 0$

$$\therefore 6x(2ax + b) = 0$$

$$\therefore x = 0 \quad \text{or} \quad x = -b/2a$$

But $P(x) \neq 0$ $\therefore$ triple root is $-b/2a$

$$\therefore P'(-b/2a) = 4a(-b/2a)^3 + 3b(-b/2a)^2 + c = 0$$

$$\therefore -\frac{b^3}{2a^2} + \frac{3b^3}{4a^2} + c = 0$$

$$b^3 + 8a^2c = 0$$

(c)(i) Since $0 \leq \theta \leq 2\pi$, $0 \leq 5\theta \leq 10\pi$,

$$\therefore 5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\therefore \theta = \pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5$$

(ii) Let $x = \cos \theta$

$$\therefore 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta + 1 = 0$$

$$\text{i.e.} \quad \cos 5\theta + 1 = 0$$

$$\therefore \theta = \pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5 \quad (\text{from above})$$

$$\therefore x = \cos \theta = \cos \pi/5, \cos 3\pi/5, \cos \pi, \cos 7\pi/5, \cos 9\pi/5$$

$$\text{and} \quad \cos 7\pi/5 = \cos 3\pi/5, \quad \cos 9\pi/5 = \cos \pi/5$$

$\therefore$ By sum of roots,

$$\cos \pi/5 + \cos 3\pi/5 - 1 + \cos 3\pi/5 + \cos \pi/5 = 0$$

$$\therefore 2(\cos \pi/5 + \cos 3\pi/5) = 1$$

$$\therefore \cos \pi/5 + \cos 3\pi/5 = +1/2$$

QUESTION 8:

$$(a) \quad x = 3 \sin \theta \Rightarrow \frac{dx}{d\theta} = 3 \cos \theta$$

$$dx = 3 \cos \theta d\theta$$

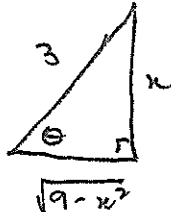
$$\therefore \int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{\sqrt{9-9 \sin^2 \theta}} \cdot 3 \cos \theta d\theta$$

$$= \int 9 \sin^2 \theta d\theta$$

$$= \frac{9}{2} \int 1 - \cos 2\theta d\theta$$

$$= \frac{9}{2} \theta - \frac{9}{4} \sin 2\theta + k$$

$$= \frac{9}{2} \theta - \frac{9}{2} \sin \theta \cos \theta + k$$

Now  (from $x = 3 \sin \theta$)
gives $\cos \theta = \frac{\sqrt{9-x^2}}{3}$

$$\therefore \text{Integral} = \frac{1}{2} \sin^{-1} \frac{x}{3} - \frac{1}{2} \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} + h$$

$$= \frac{1}{2} \sin^{-1} \frac{x}{3} - \frac{x}{6} \sqrt{9-x^2} + h$$

(b) (i)
$$\begin{cases} \alpha + \beta + \gamma = 0 \\ \alpha\beta + \alpha\gamma + \beta\gamma = 4 \\ \alpha\beta\gamma = m \end{cases}$$

$$\therefore \frac{1}{\alpha^2\beta\gamma} + \frac{1}{\alpha\beta^2\gamma} + \frac{1}{\alpha\beta\gamma^2} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{4}{m^2}$$

(ii) Since $\alpha\beta\gamma = m$
then $\alpha^2\beta\gamma = \alpha m$
 $\Rightarrow \frac{1}{\alpha^2\beta\gamma} = \frac{1}{\alpha m}$

(iii) Polynomial is $x^3 - (\text{sum of roots})x^2 + (\text{sum in pairs})x - \text{product} =$

$$\text{Sum in pairs} = \frac{1}{\alpha\beta^2\gamma^2} + \frac{1}{\alpha^2\beta^3\gamma^2} + \frac{1}{\alpha^2\beta^2\gamma^3}$$

$$= \frac{\alpha + \beta + \gamma}{\alpha^3\beta^3\gamma^3} = 0 \text{ (from above)}$$

$$\text{Product} = \frac{1}{\alpha^4\beta^4\gamma^4}$$

$$= \frac{1}{m^4}$$

$\therefore$ Polynomial is $x^3 - \frac{4}{m^2}x^2 + 0x - \frac{1}{m^4} = 0$ ← from part (i)

$$\therefore m^4 x^3 - 4m^2 x^2 - 1 = 0$$

QUESTION 9:

(a)
$$\int \frac{dx}{\sqrt{3+2x-x^2}} = \int \frac{dx}{\sqrt{3-(x-1)^2+1}}$$

$$= \int \frac{dx}{\sqrt{4-(x-1)^2}}$$

$$= \sin^{-1} \left(\frac{x-1}{2} \right) + h$$

(b) (i) Intersection of $y = \frac{1}{x}$ and $y = x^3 - 2x - 1$

$$\text{ie } x^3 - 2x - 1 = \frac{1}{x}$$

$$\therefore x^4 - 2x^2 - x - 1 = 0$$

$$\therefore x^4 - 2x^2 - x - 1 = 0$$

(ii) $P(\sqrt{x}) = (\sqrt{x})^3 + 2(\sqrt{x})^2 - (\sqrt{x}) - 1 = 0$

$$\sqrt{x}(x-1) = 1 - 2x$$

$$\therefore x(x-1)^2 = (1-2x)^2$$

$$\therefore x^3 - 2x^2 + x = 1 - 4x + 4x^2$$

$$\therefore x^3 - 4x^2 + 5x - 1 = 0$$

$$= 0$$

(iii) In above, $\alpha^2 + \beta^2 + \gamma^2 = 6$

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = 5$$

$$\alpha^2\beta^2\gamma^2 = 1$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$$

$$= 5.$$

(iv) If M is the foot of the perpendicular from A to the x-axis.

$$OA^2 = \alpha^2 + \left(\frac{1}{\alpha}\right)^2 \text{ by distance formula.}$$

$$\therefore OA^2 + OB^2 + OC^2 = \alpha^2 + \left(\frac{1}{\alpha}\right)^2 + \beta^2 + \left(\frac{1}{\beta}\right)^2 + \gamma^2 + \left(\frac{1}{\gamma}\right)^2$$

$$= \alpha^2 + \beta^2 + \gamma^2 + \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}\right)$$

$$= 6 + 5$$

$$= 11$$